

SAT Math Mastery · Week 3A

Functions & Graph Reading

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	function notation · composition · transformations of graphs and tables · reading graphs, crossings and zeros · $f(x) = k$ solution counts · domain and range · piecewise and graph-given functions · average rate of change
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Evaluate $f(a)$, $f(a + h)$, $f(kx)$, $f(g(x))$ from a formula, a table, or a graph.
- S2.** Read a graph: values $f(a)$, intercepts, maxima/minima, and the inputs that give a given output.
- S3.** Count solutions of $f(x) = k$ (and $f(x) = g(x)$) by sliding a horizontal line across the graph.
- S4.** Compose functions in either order and spot when $f(g(x)) \neq g(f(x))$.
- S5.** Transform graphs and tables: $f(x) + k$, $f(x - h)$, $af(x)$, $f(kx)$, and recover the equation of a transformed graph.
- S6.** Use symmetry and equal outputs: $f(a) = f(b)$ and $f(x) = f(c - x)$.
- S7.** Find domain restrictions (zero denominators, even roots) and the effect of a stated input interval.
- S8.** Compute average rate of change $\frac{f(b) - f(a)}{b - a}$ from a graph, table, or formula.

2 · REQUIRED FACTS

- $f(a)$ means: input a , output $f(a)$. On a graph it is the *height* of the curve above $x = a$.
- A graph passes the vertical-line test; the equation $f(x) = k$ is solved by drawing the horizontal line $y = k$ and counting intersections.
- $f(x) = g(x)$ has as many solutions as the graphs of f and g have common points; touching a curve counts once.
- Composition works inside out: $f(g(x))$ means evaluate g first, then feed that output into f .
- Replacing x by $x - h$ moves the graph h units *right*; adding k outside moves it k units *up*.
- $f(x) = f(c - x)$ for all x means the graph is symmetric about the vertical line $x = \frac{c}{2}$.
- A denominator can never be 0 and an even root needs a non-negative radicand; a stated interval can remove more inputs.
- Closed dot: endpoint included. Open dot: endpoint excluded.

3 · REQUIRED FORMULAS

$$(f \circ g)(x) = f(g(x)), \quad g(x) = f(x - h) + k \Rightarrow \text{every point } (a, b) \text{ moves to } (a + h, b + k)$$

$$\text{average rate of change on } [a, b] = \frac{f(b) - f(a)}{b - a}, \quad f(x) = f(c - x) \Rightarrow \text{axis } x = \frac{c}{2}$$

$$\text{even/odd: } f(-x) = f(x) \text{ (mirror in the } y\text{-axis)}, \quad af(x) : \text{stretches } y\text{-values by } a \text{ (reflects if } a < 0)$$

4 · MINI THEORY SHEET

Reading a function from each representation.

Given	To find $f(a)$ / solve $f(x) = k$
formula	substitute a (use parentheses around negatives); solve the equation for x
table	find the row with input a ; look for rows with output k (there may be several)
graph	go to $x = a$ and read the height; draw $y = k$ and read every crossing's x
piecewise	first decide which branch the input belongs to, then substitute

Composition from a graph or table. Do it in two reads: the inner output becomes the next input. Write the intermediate value down.

Transformations. Inside the bracket the change is horizontal and *opposite*: $f(x + 3)$ moves left 3. Outside the bracket it is vertical and *as written*: $f(x) - 2$ moves down 2. Do the inside change first when both appear.

Equal outputs. If $f(a) = f(b)$ for a function with a symmetry axis, then the axis is $x = \frac{a+b}{2}$.

Counting crossings. Count each distinct x ; a tangent contact is one point, a vertex that sits exactly on the line is one point, endpoints matter only when they are closed dots.

Function notation in context. $f(5) = 105$ is one input–output pair. $f(t + 3) - f(t)$ is the change over 3 units of input. $f(t + 1)/f(t)$ is the growth factor for exponentials.

Speed shortcuts.

Situation	Shortcut
$f(g(x))$ with numbers	work inside out and write the middle value
$f(ax + b) = \dots$ and need $f(c)$	set $ax + b = c$, solve, substitute (do not find $f(x)$ first)
$f(a) = f(b)$ (parabola, abs. value)	axis (or vertex) at $x = \frac{a+b}{2}$
average rate of change of $px^2 + qx + r$ on $[a, b]$	$p(a + b) + q$ (no need to evaluate)
$f(x) = k$ count	horizontal line $y = k$ across the graph; ignore parts outside the domain
$f(f(x)) = c$ count	solve $f(u) = c$ first, then count $f(x) = u$ for every valid u
$f(t + d)/f(t)$ for ab^t	always b^d , whatever t is
transformed graph $\rightarrow$ equation	track the key point (vertex or intercept), then match a from one other point
undefined inputs	set each denominator factor to 0

5 · GUIDED EXAMPLES

Example 1 — Composition and function notation

$f(x) = x^2 - 3x$ and $g(x) = 2x + 1$. Find $f(g(2))$ and $g(f(2))$.

$g(2) = 5$, then $f(5) = 25 - 15 = 10$. So $f(g(2)) = 10$.

$f(2) = 4 - 6 = -2$, then $g(-2) = -4 + 1 = -3$. So $g(f(2)) = -3$. Order matters.

Example 2 — Counting solutions of $f(x) = k$

The graph of $y = f(x)$ is a parabola with vertex $(1, 6)$ opening downward, on the domain $-2 \leq x \leq 4$ (closed dots), with $f(-2) = -3$ and $f(4) = -3$. How many values of x satisfy $f(x) = 0$?

The height 0 lies below the vertex (6) and above the endpoint heights (-3), so the line $y = 0$ cuts the arch twice: **2 solutions**. Had the line been $y = 6$ it would touch once; $y = -5$ misses the graph entirely.

Example 3 — Transformations and average rate of change

The point $(3, -2)$ is on $y = f(x)$. The function g is $g(x) = 2f(x - 1) + 5$. Which point is on $y = g(x)$, and what is the average rate of change of $h(x) = x^2 + 4x$ on $[1, 5]$?

$x \rightarrow x + 1$ (shift right), $y \rightarrow 2y + 5$: $(3, -2) \rightarrow (4, 2(-2) + 5) = (4, 1)$.

Shortcut: $p(a + b) + q = 1(1 + 5) + 4 = 10$. Check: $\frac{h(5) - h(1)}{4} = \frac{45 - 5}{4} = 10$.

6 · PRACTICE SET — 34 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

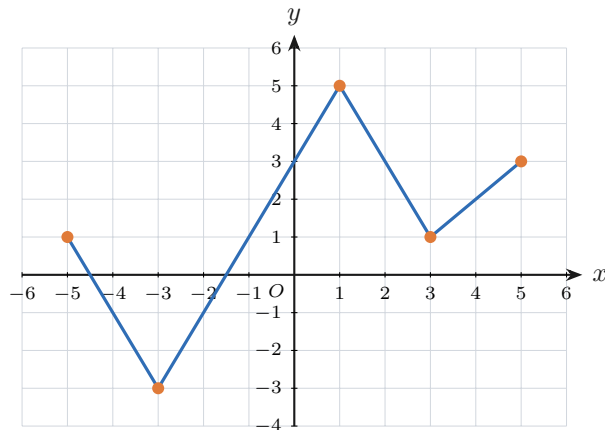
1. EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 3x^2 - 2x$. What is the value of $f(-2)$?

- (A) 8 (B) 16
(C) -16 (D) 40

2. EASY · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown in the xy -plane. What is the value of $f(0) + f(3)$?



3.

EASY · NO CALCULATOR NEEDED

The table gives values of the function f . What is the value of $f(f(2))$?

x	1	2	3	4
$f(x)$	3	4	1	2

(A) 4

(B) 1

(C) 2

(D) 3

4.

EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = x^2 - 4x$. The graph of $y = g(x)$ is the graph of $y = f(x)$ translated 3 units up. Which of the following defines g ?

(A) $g(x) = x^2 - 4x + 3$ (B) $g(x) = x^2 - 4x - 3$ (C) $g(x) = (x + 3)^2 - 4(x + 3)$ (D) $g(x) = (x - 3)^2 - 4(x - 3)$ **5.**

EASY · NO CALCULATOR NEEDED

The function g is defined by $g(x) = \frac{12}{x - 3}$. For which of the following values of x is $g(x)$ undefined?

(A) 12

(B) -3

(C) 0

(D) 3

6. MEDIUM · NO CALCULATOR NEEDED

Functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(2)) - g(f(2))$?

7. MEDIUM · NO CALCULATOR NEEDED

The function f is defined by $f(x) = (x - 2)(x + 6)$. If $f(m) = f(-3)$ and $m \neq -3$, what is the value of m ?

8. MEDIUM · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 3x^2 - x$. Which of the following expressions is equivalent to $f(2x)$?

(A) $12x^2 - x$

(B) $6x^2 - x$

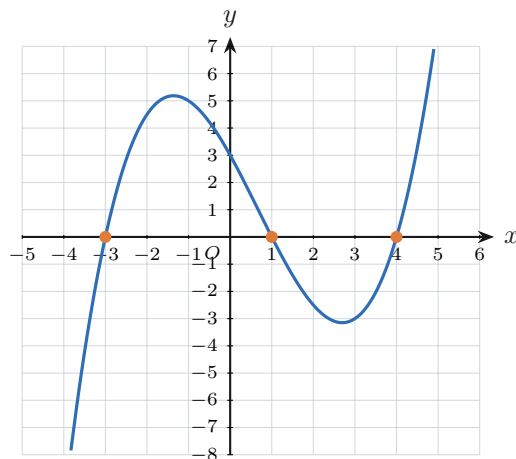
(C) $12x^2 - 2x$

(D) $6x^2 - 2x$

9.

MEDIUM · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown in the xy -plane. For how many values of x does $f(x) = 2$?



(A) 2

(B) 3

(C) 4

(D) 1

10.

MEDIUM · NO CALCULATOR NEEDED

The table shows values of the function f . The function g is defined by $g(x) = f(x - 2) + 1$. What is the value of $g(5)$?

x	1	3	5	7
$f(x)$	4	-2	6	9

(A) 10

(B) -3

(C) 7

(D) -1

11.

MEDIUM · CALCULATOR

The function $p(n) = 18n - 240$ models the profit, in dollars, from selling n tickets to a school fundraiser. What is the least whole number of tickets that must be sold so that the profit is positive?

12.

MEDIUM · NO CALCULATOR NEEDED

The function f is defined by $f(x) = \begin{cases} 2x + 1, & x < 3 \\ x^2 - 4, & x \geq 3 \end{cases}$. What is the value of $f(f(1))$?

(A) 5

(B) 7

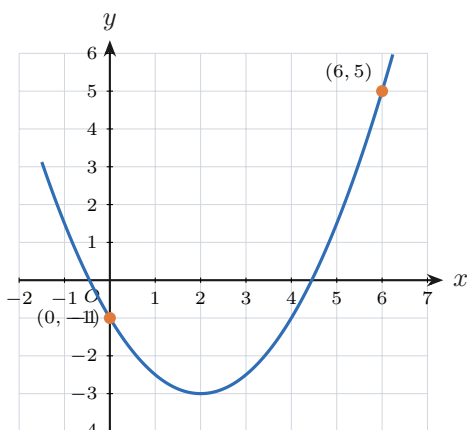
(C) 3

(D) -5

13.

MEDIUM · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown, with two points labeled. What is the average rate of change of f from $x = 0$ to $x = 6$?



(A) 5

(B) 1

(C) 2

(D) 6

14.

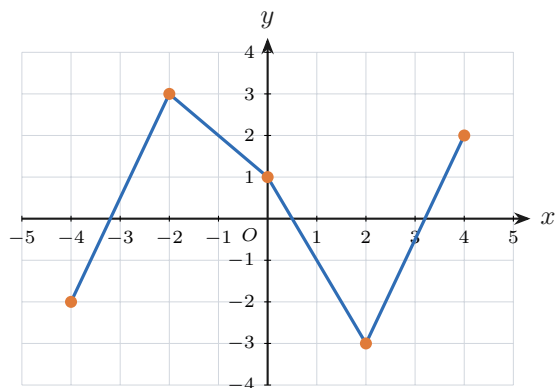
HARD · NO CALCULATOR NEEDED

Functions f and g are defined by $f(x) = 2x - 1$ and $g(x) = x^2 + 2$. There are two values of a for which $f(g(a)) = g(f(a))$. What is the positive one?

15.

HARD · NO CALCULATOR NEEDED

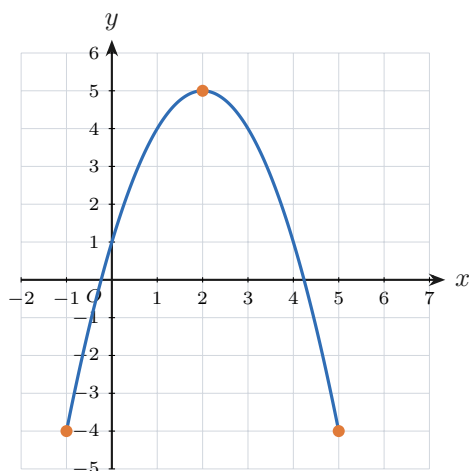
The graph of the piecewise linear function f is shown, with the vertices at grid points. What is the value of $f(f(0)) + f(f(1))$?



16.

HARD · DESMOS

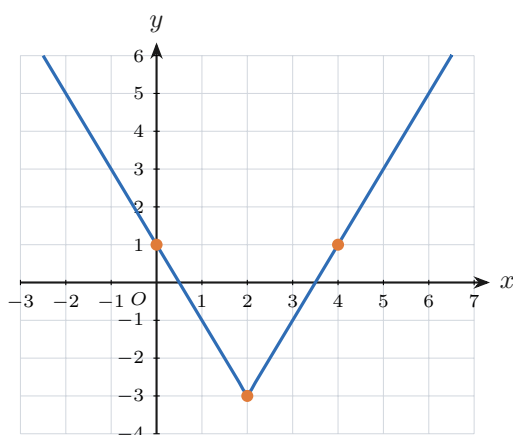
The graph of $y = f(x)$ is shown. It is a portion of a parabola on the interval $-1 \leq x \leq 5$, and the endpoints are closed dots. For how many integer values of k does the equation $f(x) = k$ have exactly two solutions?



17.

HARD · NO CALCULATOR NEEDED

The graph of $y = g(x)$ is shown. Which of the following defines g ?



(A) $g(x) = 4|x - 2| - 3$

(B) $g(x) = 2|x - 2| + 3$

(C) $g(x) = 2|x - 2| - 3$

(D) $g(x) = 2|x + 2| - 3$

18.

HARD · NO CALCULATOR NEEDED

The function g is defined by $g(x) = 3 \cdot 2^x$. For what value of a is $g(a + 2) = g(a) + 36$?

(A) 2

(B) 3

(C) 4

(D) 1

19.

HARD · NO CALCULATOR NEEDED

The function f satisfies $f(3x - 2) = x^2 + 5$ for all values of x . What is the value of $f(10)$?

(A) 105

(B) 36

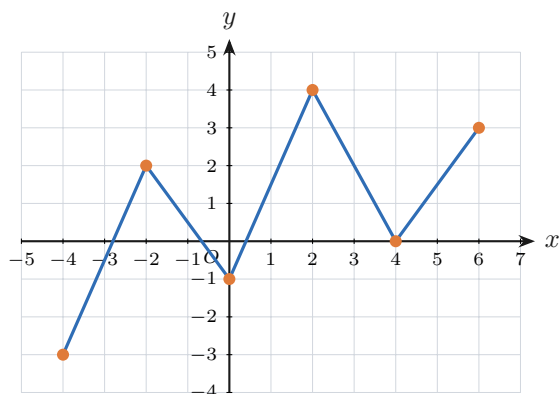
(C) 9

(D) 21

20.

HARD · DESMOS

The graph of the piecewise linear function f is shown for $-4 \leq x \leq 6$. For how many values of x in this interval does $f(x) = 1$?



(A) 3

(B) 4

(C) 5

(D) 6

21.

HARD · NO CALCULATOR NEEDED

The table shows values of the function f . The function g is defined by $g(x) = 3f(x + 1) - 2$. What is the value of $g(-1) + g(3)$?

x	0	2	4	6
$f(x)$	-1	3	7	2

(A) 6

(B) 14

(C) 18

(D) 22

22.

HARD · CALCULATOR

The function $V(t) = 24000(0.85)^t$ gives the value, in dollars, of a delivery van t years after it was bought. Which of the following statements must be true?

I. $V(t + 1) = 0.85 V(t)$ for every t .

II. $V(t + 2) = 0.70 V(t)$ for every t .

III. The van is worth less than \$12,000 when $t = 5$.

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only

23.

HARD · NO CALCULATOR NEEDED

The function g is defined by $g(x) = \frac{1}{x^2 - 6x + k}$, where k is a constant. The function is undefined for exactly one value of x . What is the value of k ?

(A) 9

(B) 12

(C) 3

(D) 6

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

The linear function f satisfies $f(f(x)) = 9x + 8$ for all x , and $f(1) = 5$. What is the value of $f(10)$?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

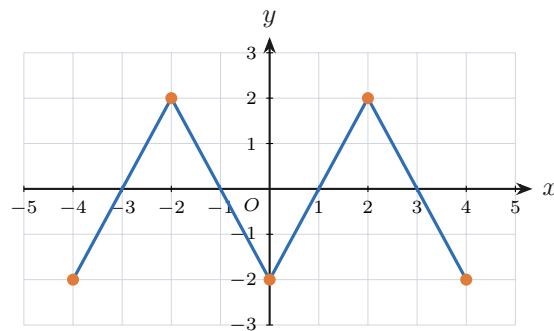
The function f satisfies $f(x) = f(8 - x)$ for all real x , and $g(x) = f(x + 4)$. If $g(3) = 11$ and $g(-1) = 5$, what is the value of $f(5) + f(7)$?

- (A) 10 (B) 16
(C) 21 (D) 22

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of the piecewise linear function f is shown for $-4 \leq x \leq 4$. For how many values of x in this interval does $f(f(x)) = 0$?



27.

SAT CHALLENGE · DESMOS

The function f is defined by $f(x) = -2|x - 1| + 8$. The graph of $y = f(x + 3) - 2$ and the line $y = k$ intersect at two points that are 9 units apart. What is the value of k ?

- (A) 3
- (B) 15
- (C) -3
- (D) -1

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

The functions f and g are defined only for $x = 1, 2, 3, 4, 5$, and their values are in the table. What is the sum of all values of x for which $f(g(x)) = g(f(x))$?

x	1	2	3	4	5
$f(x)$	3	5	1	2	4
$g(x)$	2	3	5	1	4

29.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = ax^2 + 4x$, where a is a constant. The average rate of change of f over the interval $1 \leq x \leq 5$ is 22. What is the value of $f(3)$?

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = \frac{ax+5}{x-3}$ for $x \neq 3$, where a is a constant. If $f(f(x)) = x$ for every x in the domain of $f \circ f$, what is the value of $f(7)$?

(A) $\frac{13}{2}$

(B) $\frac{19}{4}$

(C) 26

(D) -4

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

For $x \geq 0$, let $f(x) = 120 - 90(0.8)^x$ and $g(x) = -3(x - 4)^2 + 125$. Which of the following statements is true?

- (A) Neither attains a maximum
(B) Both f and g attain a maximum
(C) Only f attains a maximum, 120
(D) Only g attains a maximum, 125

32.

SAT CHALLENGE · DESMOS

$$2^x = 3x + k$$

In the equation above, k is a constant. What is the smallest integer value of k for which the equation has two distinct real solutions?

- (A) 0
(B) -3
(C) -2
(D) -1

33.

SAT CHALLENGE · DESMOS

Let $g(x) = \log_2 x$ for $x > 0$ and $h(x) = \frac{x+8}{6}$. For which values of x is $h(x) > g(x)$?

- (A) $x < 4$ or $x > 16$ (B) $0 < x < 4$ or $x > 16$
(C) $4 < x < 16$ (D) $x > 16$ only

34.

SAT CHALLENGE • NO CALCULATOR NEEDED

The table gives values of a quadratic function g at equally spaced inputs.

x	2	5	8	11
$g(x)$	44	95	200	359

What is the value of $g(17)$?

- (A) 677 (B) 572
(C) 785 (D) 839